Answer all the questions

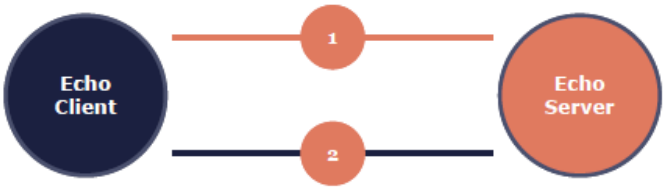
1- Write the main **components** of distributed system.

2- Define the following:
   a- **Nodes** and **processes** with illustrating the **types** of each one.
   b- **Communication link** with describing of communication **Characteristics.**
   c- **Resource Sharing** and its **Mechanisms**.

3- Write the reasons of using a **socket** and **port number** and what are the **required classes** in java program for creating a socket for client and server program.

4- Write the **required steps** for implementing a **server** and **client** program using a socket.

5- According to the following figure, assume you have two nodes (node A and node B) **node A** need to send a requesting message "**are u ready**" to node B and **node B** response to node A with a message "**yes I'm ready**",



Write a simple program in java to illustrate how a node A (client) and node B (server) communicate through TCP sockets (*consider host name Localhost, portnumber 6000*).

I hope you answer all the questions with my best wishes